Xuehai Zhou

McGill University
Department of Bioresource Engineering
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Research Fields

Artificial intelligence; Plant phenotyping; Radiometric imaging; Sensor fusion; 3D point cloud processing; Feature extraction

Education

McGill University — **Montreal, QC**
Ph.D. Student in Bioresource Engineering — Expected: Dec 2027
Advisor: Prof. Shangpeng Sun
Committee Member: Prof. Pierre Dutilleul

McGill University — **Montreal, QC**
M.Sc. in Bioresource Engineering — December 2023
Advisor: Prof. Shangpeng Sun
Committee Member: Prof. Mark Lefsrud

University of Oregon — **Eugene, OR**
B.Sc. in Computer Science — June 2021

Appointments

Graduate Researcher — **Dec. 2022 – Present**
Department of Bioresource Engineering — McGill University

Research Assistant — **May 2021 – Oct. 2022**
Baidu Research — Beijing, China

Tutor — **Mar. 2020 - Mar. 2021**
Department of Computer Science — University of Oregon

Paper Marker — **Mar. 2019 – Mar. 2020**
Department of Mathematics — University of Oregon

Publications

Journal

5. **Xuehai Zhou**, Tianzi Yang, Rui Xu, Alexander Bucksch, Pierre Dutilleul, Davoud Torkamaneh, Shangpeng Sun. *"3D skeletonization and phenotyping for soybean root system architecture using a bio-inspired algorithm."* Computers and Electronics in Agriculture, 239, 110890, 2025.

4. Zhouzhou Zheng, Mohamed Debbagh, **Xuehai Zhou**, Shangpeng Sun, Yuxiang Huang. *"Graph-Transformer with spatial-spectral features fusion for hyperspectral image classification."* Expert Systems with Applications, 264, 125962, 2025.

3. **Xuehai Zhou**, Yuyang Zhang, Xintong Jiang, Kashif Riaz, Phil Rosenbaum, Mark Lefsrud, Shangpeng Sun. *"Advancing tracking-by-detection with MultiMap: Towards occlusion-resilient online multiclass strawberry counting."* Expert Systems with Applications, 255, 124587, 2024.

2. Rui Kang, Jiaxin Huang, **Xuehai Zhou**, Ni Ren, Shangpeng Sun. *"Toward real scenery: A lightweight tomato growth inspection algorithm for leaf disease detection and fruit counting."* Plant Phenomics, 6, 0174, 2024.

1. Ji Liu, **Xuehai Zhou**, Lei Mo, Shilei Ji, Yuan Liao, Zheng Li, Qin Gu, and Dejing Dou. *"Distributed and deep vertical federated learning with big data."* Concurrency and Computation: Practice and Experience, e7697, 2023.

Conference

3. **Xuehai Zhou**, Marc-Antoine Chiasson, Liwen Han, Davoud Torkamaneh, Pierre Dutilleul, Shangpeng Sun. *"A high-throughput pipeline for 3D reconstruction of soybean root system architecture using macro-CT scanning."* ASABE Annual International Meeting, 2025.

2. **Xuehai Zhou**, Leshang Bai, Rui Xu, Rui Kang, Davoud Torkamaneh, Shangpeng Sun. *"3D segmentation within the root system architecture using Point Transformer."* ASABE Annual International Meeting, 2024.

1. **Xuehai Zhou**, Yuyang Zhang, Shangpeng Sun, Phil Rosenbaum. *"A dynamic object counting method for strawberry fruits using vision transformer networks and Kalman filter tracking."* ASABE Annual International Meeting, 2023.

Awards

FRQNT Doctoral Training Scholarships — **2025-2029**
The Fonds de recherche du Québec – Nature et technologies (FRQNT) — CA$91,667

Graduate Excellence Award — **2025**
McGill University — CA$20,000

Student Paper Award — **2023**
Association of Overseas Chinese Agricultural, Biological, and Food Engineers

ITSC Paper Award — **2023**
American Society of Agricultural and Biological Engineers

Dean's List — **2018**
University of Oregon

Presentations

ASABE Annual International Meeting, 2025 — **Oral Lightning**
Title: *3D reconstruction of root system using macro-CT scanning* — Toronto, Ontario

4e Congrès annuel du RQRAD, 2025 — **Oral Lightning**
Title: *Collaborative mobile system for precision spraying* — Quebec City, Quebec

ASABE Annual International Meeting, 2024 — **Oral Lightning**
Title: *3D segmentation within the root system architecture* — Anaheim, California

ASABE Annual International Meeting, 2023 — **Oral Presentation**
Title: *A dynamic object counting method for strawberry fruits* — Omaha, Nebraska

Paper review

Computers and Electronics in Agriculture

Biosystems Engineering

Journal of the ASABE

Plant Phenomics

Plant Methods

Signal, Image and Video Processing

Scientific Reports

Smart Agricultural Technology